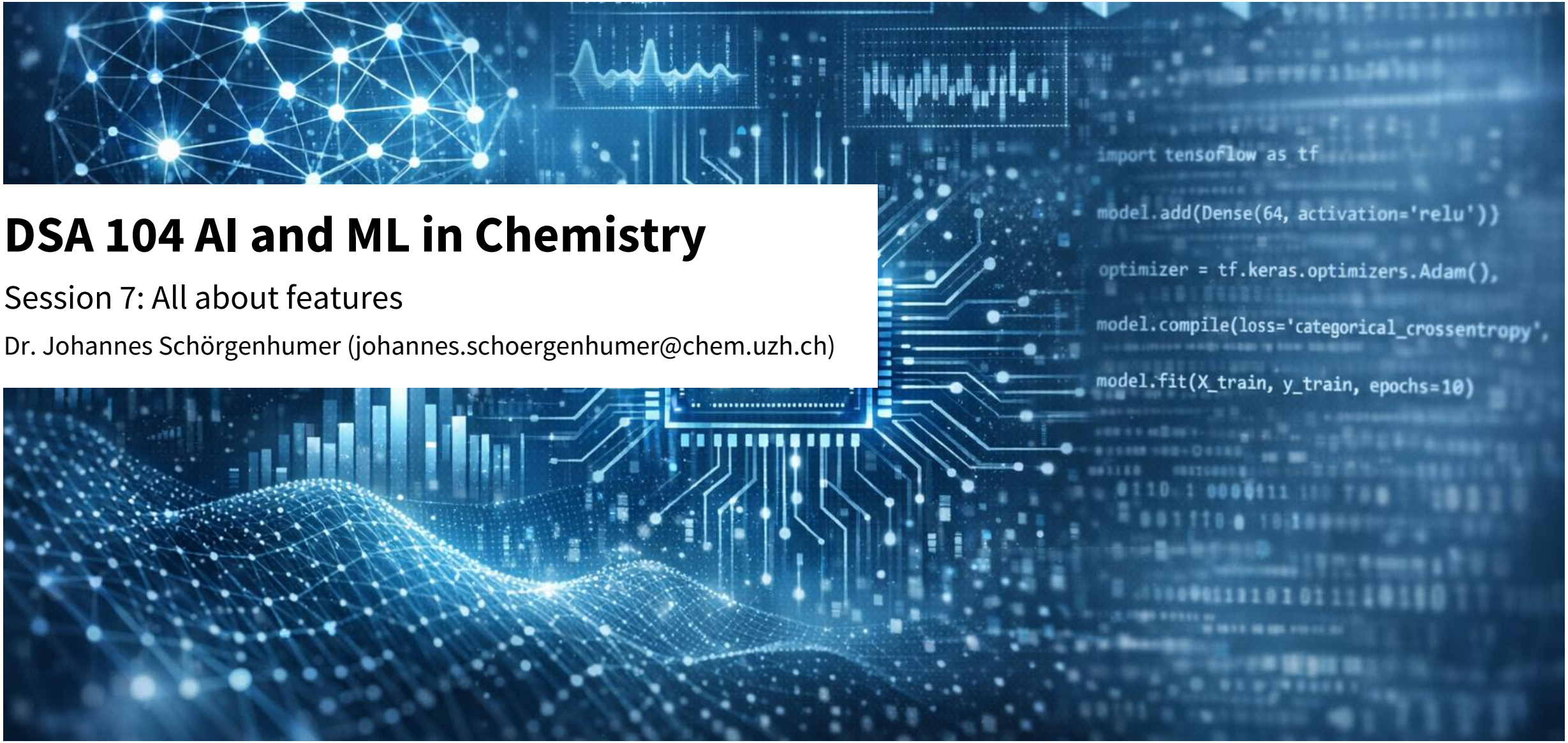
Universität Zürich[UZH]

Department of Chemistry



DSA 104 AI and ML in Chemistry

Session 7: All about features

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

Lesson outline

Wrap up Unsupervised Learning

Feature Engineering

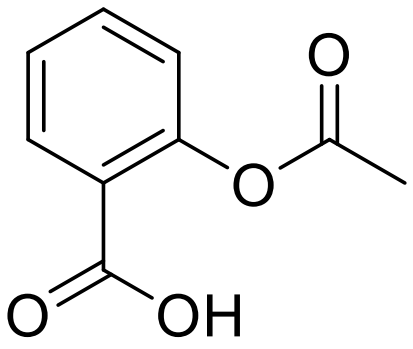
Feature Selection

Clustering: Exploring chemical space

Use the modified ESOL dataset (*esol_modified.csv*) to explore the physicochemical diversity.

1) Data preprocessing
2) Dimensionality reduction: compare PCA, t-SNE, UMAP in visualisation
3) Try KMeans and DBSCAN as clustering methods
4) Evaluate the clustering

Why features matter



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>MCHU - Calmodulin - Human, rabbit,
bovine, rat, and chicken
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NPTEAELQDMINEVDADGNGTID
FPEFLTMMARKMKDTDSEEEIREAFRVFDKDGNGYISAAELR
HVMTNLGEKLTDEEVDEMIREA
DIDGDGQVNYEEFVQMMTAK*
```

How can we teach a model to “understand” a molecule or a protein sequence?

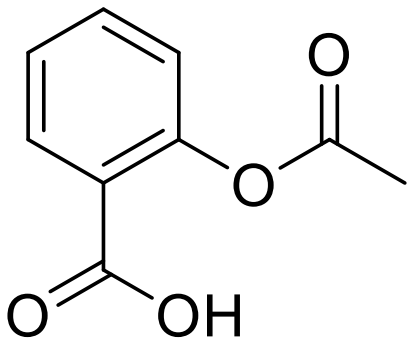
— ML models are only statistical machines!

— ML models need numbers!

— Feature engineering = translating chemistry into clean, numeric representation

— Model quality mostly limited by features

How can we encode reality?



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```

Molecules, e.g.:

— molecular weight

— logP

— polar surface area

— hydrogen bond donors/acceptors

— rotatable bonds

— SMILES

— 3D conformation…

Proteins, e.g.:

— Amino acid composition

— Hydrophobicity index

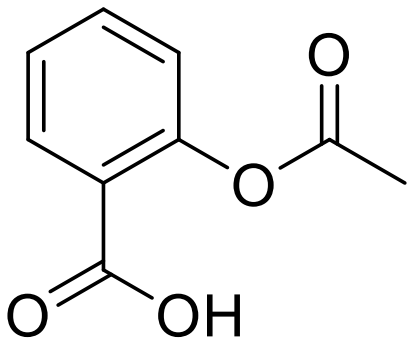
— Binding sites

— Charge

— Motif counts

— k-mer frequencies

How can we encode reality?



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```

— Physicochemical descriptors (mainly numerical)

— Categorical features

— Molecular structure encoding, e.g. binary.

Need to hit the sweet spot for the models! (Quality and quantity of features!)

Feature engineering!

— Too many features → overfitting

— Wrong features → model learns nothing

— Well engineered features → huge performance gains

Feature engineering: the bread and butter of ML

Domain knowledge, data driven, synthetic

Data cleaning
Normalisation
Math. Transformations
Encoding

E.g. Dimensionality reduction

Filter methods
Wrapper methods
Embedded methods

Min-max
Standard scaler

https://www.geeksforgeeks.org/machine-learning/what-is-feature-engineering/

Feature engineering: one-hot encoding

Qualitative / **categorical data problematic** for models (numerical methods fail):

— Atom type

— Amino acid

— Identity

— Functional group class

— …

— Text (experimental descriptions!)

Solution: **Transform into binary format**!

```
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MADQLTEEQIAEFKEAFSLFDKDGDGTITTKELGTVMRSLGQ
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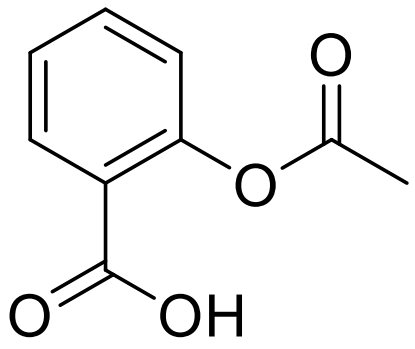

Feature engineering: one-hot encoding

- Create **vocabulary** / dictionary for n categories
- Describing each value as integer not feasible
- Hence: Represent each feature as n-dimensional, **one-hot-encoded** feature vector: $v = (v_1, v_2, \dots v_v)$:
  - Data with *n* different values enumerated from 1…*n* in **fixed** order
  - $v_i = 1$ if category *i* is true, otherwise $v_i = 0$
  - Each element represents one category, **no false information or bias**
- Implementation:
  - Pandas: get_dummies()
  - Scikit-Learn: OneHotEncoder

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HVMTNLGEKLTDEEVDEMIREA
DIDGDGQVNYEEFVQMMTAK*
```

~~"M" = 1, "A" = 2, "D" = 3, …~~

Vocabulary:

Amino acid	A	R	N	D	C	…
Position	1	2	3	4	5	

$$v_{\mathrm{A}} = (1, 0, 0, 0, 0, \dots)$$
$$v_{\mathrm{D}} = (0, 0, 0, 1, 0, \dots)$$

Feature engineering: one-hot encoding

Pros:

— Algorithm **compatibility**!

— **Eliminate Ordinality** (possible bias due to mistaken ranking)

— Can **improve model performance** (more values instead of single entities)

— Can be concatenated with other features (hybrid encoding: richer space than just identity)

Cons:

— Vocabulary of size *n* yields vectors with *n* dimensions

— Sparse: **Large vocabularies not feasible** for rich categories (see example: 20 amino acids)

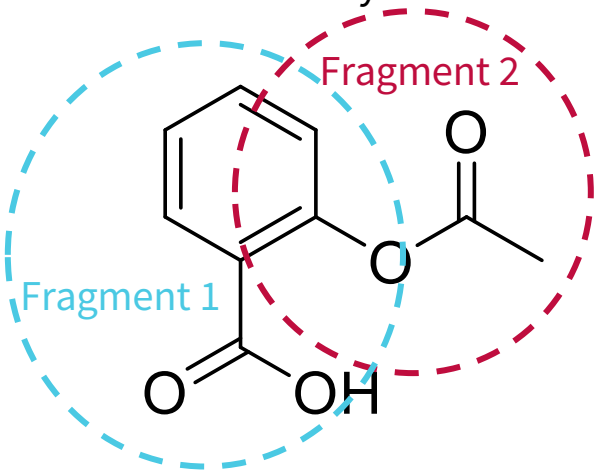
— No similarity information

…use embedded layers, e.g. word embedding (NLP, fingerprints!

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
D	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Q	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
G	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
H	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
I	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
L	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
K	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
M	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
F	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
T	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
Y	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
V	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Feature engineering: fingerprints

Encode structure as binary vector!



	Amine	Aromatic ring	Carbonyl	Hydroxy
Fragment 1	0	1	1	1
Fragment 2	0	1	1	0

Extended-Connectivity Fingerprints (ECFP / Morgan fingerprints):

— Local neighborhoods hashed into bit vectors

— Typically: 1024 or 2048 bits

— Well-suited for capturing structural information

— Widely applied in QSAR studies

Fingerprint generators in RDKit:
— **AtomPairFP** = rdkit.Chem.rdFingerprintGenerator.FPType.AtomPairFP
— **MorganFP** = rdkit.Chem.rdFingerprintGenerator.FPType.MorganFP
— **RDKitFP** = rdkit.Chem.rdFingerprintGenerator.FPType.RDKitFP
— **TopologicalTorsionFP** = rdkit.Chem.rdFingerprintGenerator.FPType.TopologicalTorsionFP

https://www.rdkit.org/docs/source/rdkit.Chem.rdFingerprintGenerator.html

Feature engineering: beyond structure

Physicochemical descriptors, e.g.

— Molecular weight

— logP

— Polar surface area

— Number of hydrogen bond donors

— Partial charges

Despite related to structure **often outperform structural descriptors in QSAR**

Can be concatenated to e.g. one-hot-vocabularies

Exercise: Feature engineering

1) Walkthrough: One-hot encoding using scikit-learn

2) Imagine caffeine as model compound. Derive
   1) Some meaningful one-hot encoding for the structure
   2) Come up with your own fingerprint system

3) Consider the following scenario:

You are investigating enzyme catalysis. Your dataset includes:

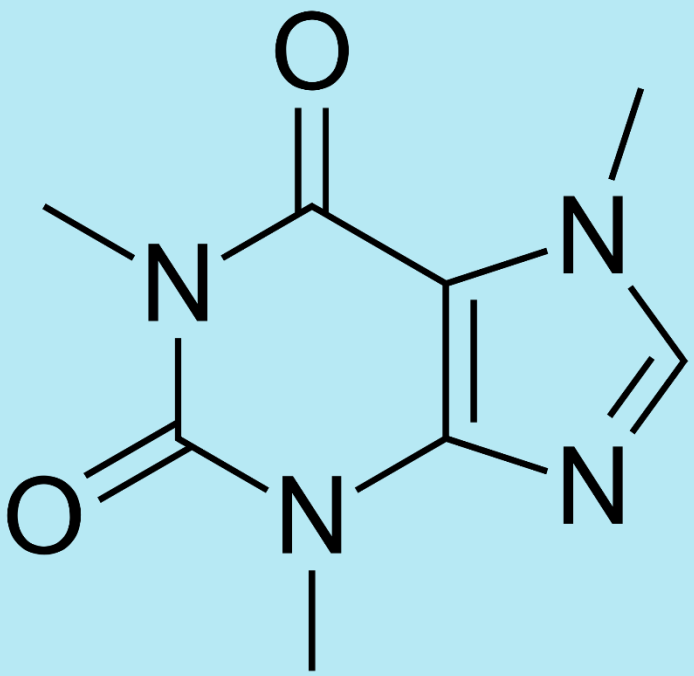
— amino acid sequence

— pH

— Temperature

— substrate structure

Discuss in small groups which features you would engineer. How and why?



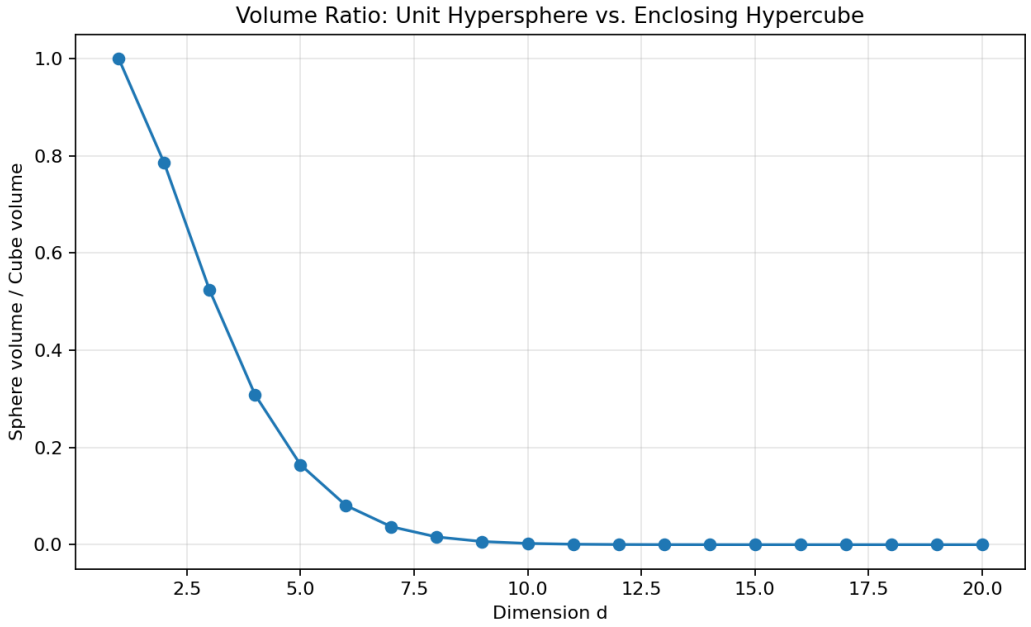
Feature selection

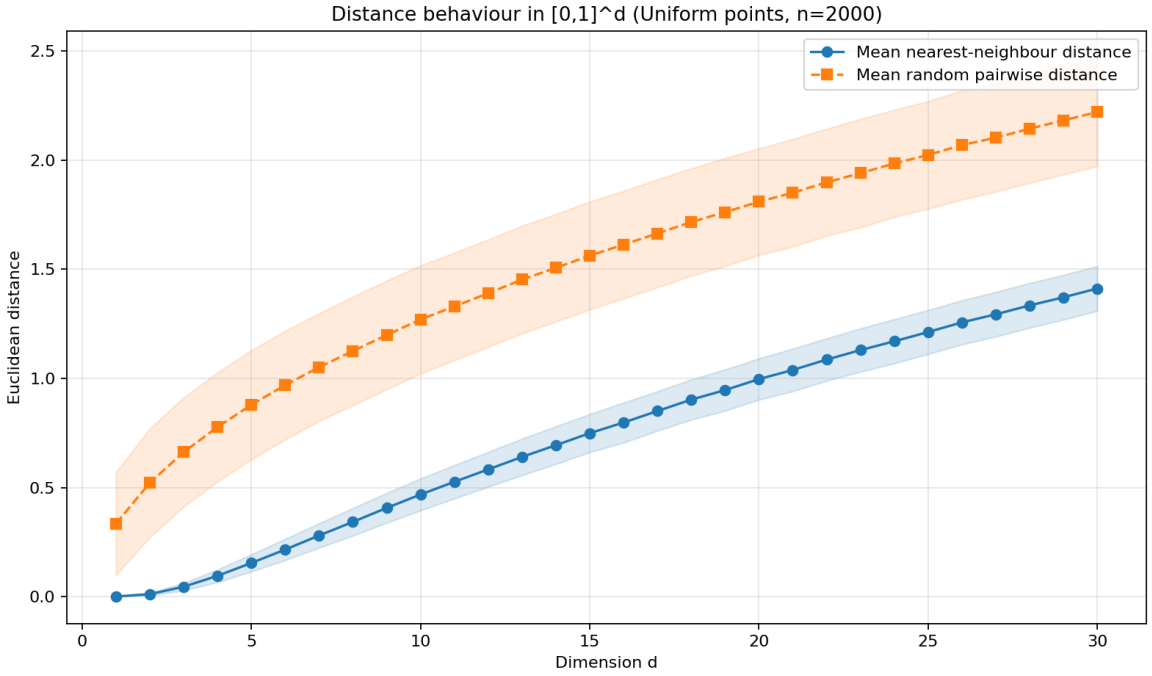
The **curse of dimensionality** (Richard E. Bellman, 1957)

- Issue: Volume grows faster than the number of data points
- Consequences:
  - Space is **mostly empty**
  - **Distance** metrics become **irrelevant**
  - Methods based on **neighbors become irrelevant**
- Effects on data analysis:
  - Amount of data **grows exponentially** with dimensionality
  - **Long training duration**
  - **Overfitting** (mostly trained on noise)

E.g. Fingerprints in a small chemistry dataset

- 2048 bit
- 100 molecules





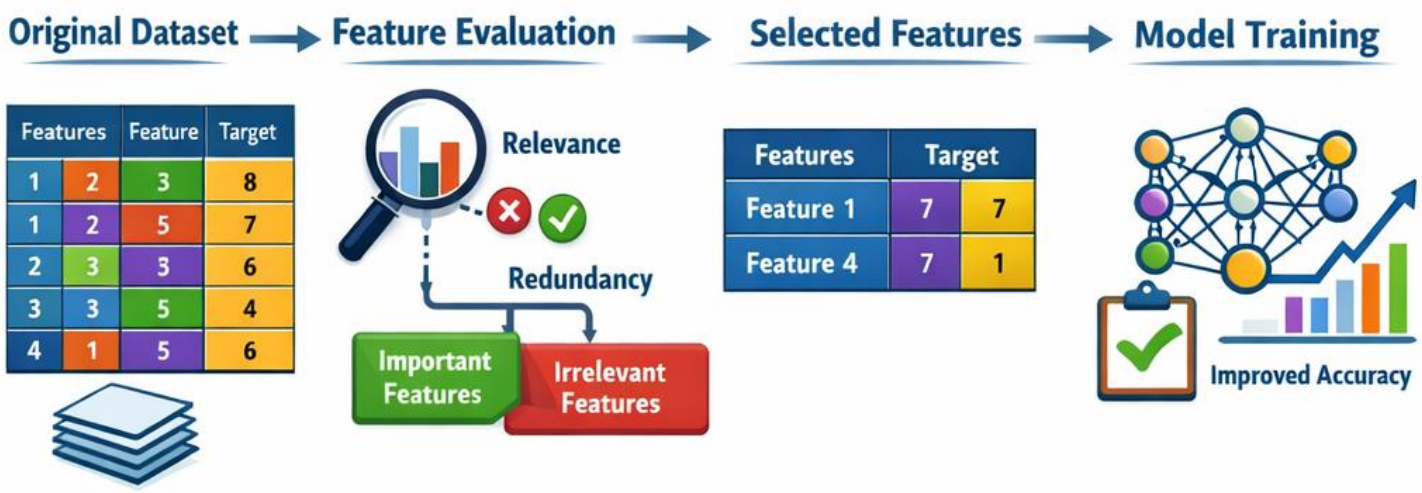
Feature engineering: Feature selection

Reduce numbers of the features needed to

- **Remove noise**
- **Reduce overfitting**
- **Improve interpretability**
- **Faster models**

How?

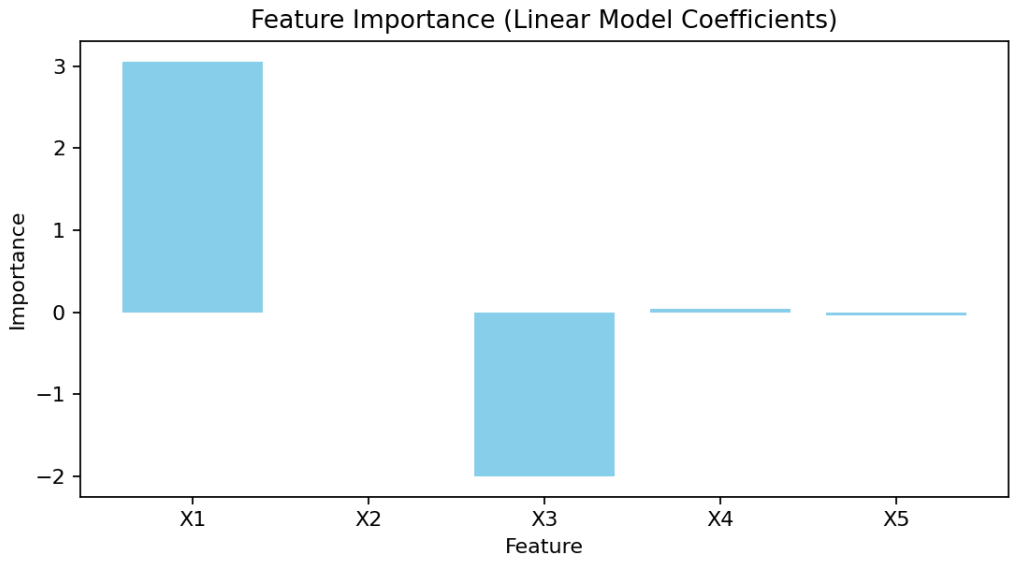
- Dimensionality reduction (PCA, TSNA)
- **Feature selection:**
  - Filter methods
  - Wrapper methods
  - Embedded methods

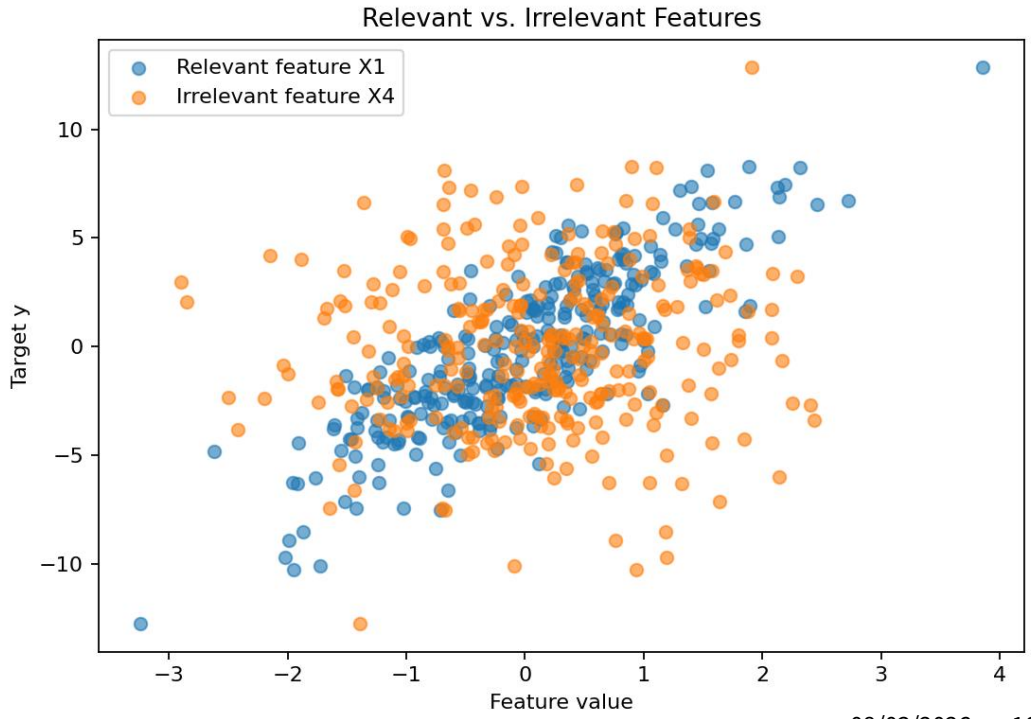


Feature selection: Filter methods

Examples:

- **Variance threshold**:
  - **Useless if no variance** encoded (e.g. fingerprint)
  - Extremely rare fragments might not matter too much (**?**)
  - Extremely common fragments removed as well
- **Correlation filtering**:
  - E.g. ESOL dataset: logP and log S
  - Again: no variance encoded!
- **Mutual information**:
  - Remove features not contributing to the prediction
  - measures **how much knowing a feature reduces uncertainty about the label**.
  - Irrelevant features removed





Feature selection: Wrapper methods

Pro: Filtering compares features individually, wrapping together (often better!)

Cons: **Expensive!**

Model repeatedly evaluates feature subsets to see how well the prediction works:

1. choose a subset of features
2. train a model
3. evaluate performance (e.g. cross-validation accuracy)
4. change the subset
5. Repeat

The algorithm searches for the feature subset that produces the **best model performance**.

Different methods:

— Forward feature selection: add one by one

— Backward elimination: reverse process

— **Recursive feature elimination**: select features based on importance (SVM, RF!)

Feature selection: Embedded methods

Model performs selection internally**. Feature selection built into the model training process.**

Pros:

— usually **better than simple filters**

— much **faster than wrapper methods**

Cons:

— very **model specific!!**

Example:

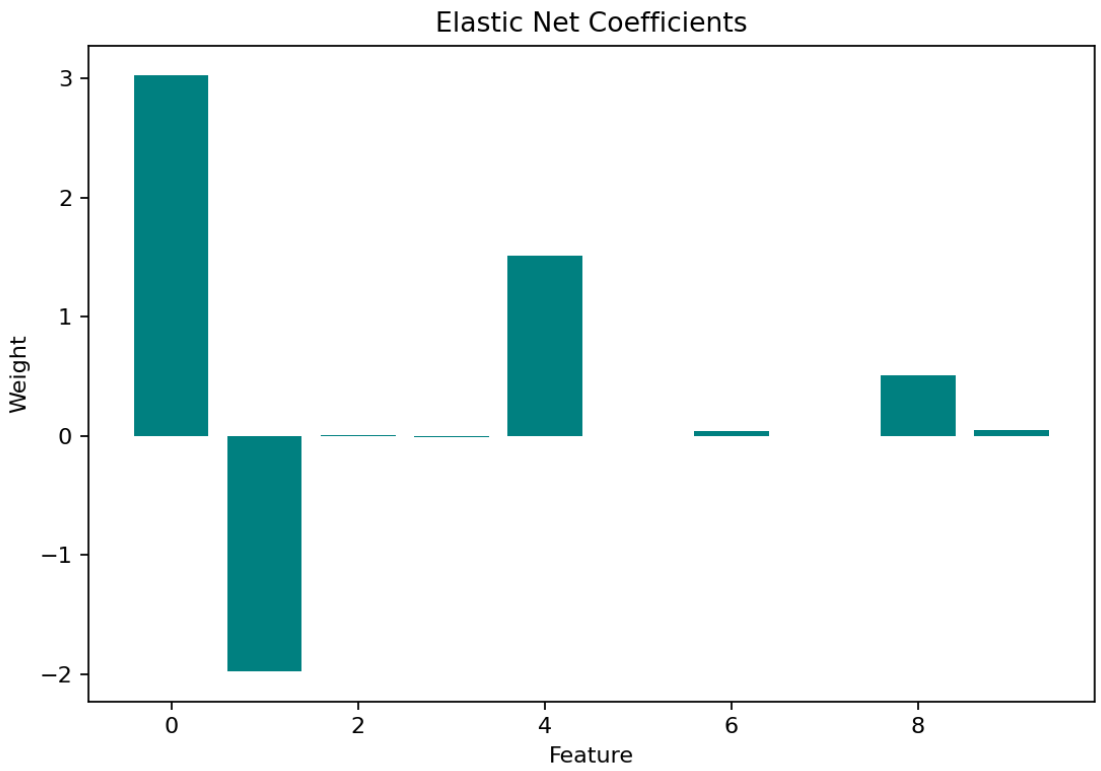
— L1 regularization (LASSO)

  — Penalty on large number of coefficients

  — Features with coefficient 0 automatically removed

— tree-based importance (RF):

  — Only useful features used in splits



Feature selection: Automatic generation vs. domain knowledge

Automatic feature generation:

- Examples:
  - fingerprints
  - learned representations

Pros:

- scalable
- captures complex structure

Cons:

- less interpretable

Domain knowledge

- Expert chooses features, e.g. molecular descriptors

Pros:

- interpretable
- robust for small datasets

Cons:

- may miss hidden patterns

Exercise: Feature selection

Scenario: You are investigating the antibacterial activity of some small compounds. Your dataset is a combination of different sources and has the following features:

PubChemID
SMILES
Antibacterial activity
Molecular weight
LogP
Solubility
TPSA
Partial charges
Heavy atom count
Toxicity
Inhibitory concentrations on cancer cells
Polarity
Count N, C, S
Price (commercial compounds)
Supplier

Discuss which features you would keep for model training. Which ones would you additionally need to engineer?